

# Benefits & Risks Evaluation Modeling

Dynamic evaluation of a drug's  
potential, assisted by AI



# ArcaScience at a Glance

Creation date: 2018

Total amount raised:  
**\$10M**

Last fundraising: **\$7M**  
in September 2025

Awarded best healthtech NLP  
Startup 2023<sup>1</sup>

**20+** blue chip clients running our solution

Empowering pharmaceutical  
professionals and regulatory  
authority bodies with  
**comprehensive, reliable**  
**BRA data** for  
better real-time decisions

<sup>1</sup>: <https://www.startus-insights.com/innovators-guide/natural-language-processing-startups/>

# ArcaScience Dramatically Changes the way these Evaluations are Conducted

## Build your drugs' BRE in seconds

*Don't spend months looking for it*

By far the most  
**comprehensive** BRE  
database in the world

+

**24** small AI models  
trained by clinicians

*Unmatched, with over 100bn data points*

...



## Delivered on a **user-friendly & transparent** solution dashboard

Benefits & Risks  
strengths, weaknesses &  
**opportunities**

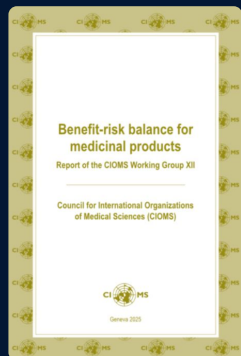


Approval **optimization**  
*Risk mitigation, new design  
amendment strategies,  
better applications*

**Reduce BRA Project Time & Study Generation by 80%<sup>1</sup>**

<sup>1</sup>: independent BRA experts evaluation, 2025

# 2025: A Transformative Year for Benefit-Risk Analysis

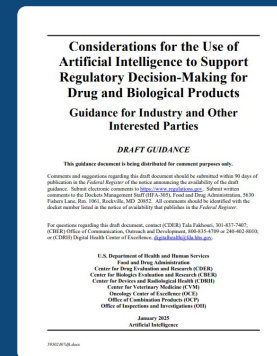


*Benefit-Risk Balance for Medicinal Products - Working Group XII*



**Regulatory Framework Development:**  
CIOMS Working Group XII releases updated guidance on benefit-risk balance methodology for medicinal products

**AI-Powered Transformation:**  
FDA publishes framework for evaluating AI model credibility and validation in drug development



*Framework to Advance Credibility of AI Models Used for Drug Development*

# Current Obstacles in Pharmaceutical Benefit-Risk Evaluation

1

Need for  
comprehensive  
integration of  
medical evidence

2

Overly long  
analysis duration

3

Need to increase  
characterisation  
in risk assessment

4

Challenges in  
real-time  
evidence  
integration

5

Insufficient  
integration of  
patient  
preferences

Innovative Approaches are Transforming Benefit-Risk Analysis

# Automated Studies Generation that Fit CIOMS Framework

## Product Opportunities

- ✓ Analysis of disease/condition
- ✓ Current treatment options

## Product Profile

- ✓ Key Benefits & Risks
- ✓ Uncertainties for Benefits & Risks

## Product Optimisation

Risk Management  
(in progress)

✓ **Benefit-Risk Conclusion**

Trusted by Leading Pharma Groups:

AstraZeneca

sanofi



✓ : Partial coverage

✓ : Available studies

# BRAD Crosswalk to External Documents

| BRAD                        | eCTD / Clinical Overview        | PBRER             |
|-----------------------------|---------------------------------|-------------------|
| ✓ Analysis of Condition     | 2.5.6.1 Therapeutic Context     | -                 |
| ✓ Current Treatment Options | 2.5.6.1 Therapeutic Context     | -                 |
| ✓ Benefits                  | 2.5.6.2 Benefits                | 18.2 Benefits     |
| ✓ Risks                     | 2.5.6.3 Risks                   | 18.2 Risks        |
| Risk Management             | 2.5.6.3 Risks                   | Add to Conclusion |
| ✓ Benefit-Risk Assessment   | 2.5.6.4 Benefit-Risk Assessment | 18.2              |
| ✓ Conclusion                | 2.5.6.4 Benefit-Risk Assessment | 18.2              |

Standardized Studies Meet Multiple Regulatory Requirements

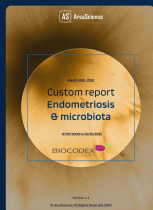
✓: Partial coverage

✓: Available Studies



# Choose from Six Pre-Configured Study Types

## Analysis of condition



### Disease Analysis

Discover molecular pathways and epidemiological information about your disease

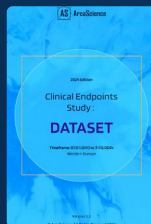
## Current Treatments



### Clinical Landscape + Efficacy Report

Access to a global synthesis of clinical activities with sponsors, treatments...

## Benefits



### Clinical Endpoint Study

Access to a accurate analysis of all clinical endpoint strategy for a pathology

## Risks



### Adverse Events Reports

Get a synthesis of all adverse events related to your drug and the standard of care

## Benefit-Risk Assessment



### Benefit Risk insight

Access to a summarized BRA regarding your priorities

## Conclusion



### Benefit Risk Summary

Review a synthesis of your tailored benefit-risk analysis

Each Study Delivers Full Data Transparency and FDA-Compliant Documentation



# Studies Alone Falls Short in Two Critical Areas

Agility in the BRA Process (Test & Learn)



Share Real-Time Updates with Stakeholders

# AI-Powered BRA Automation: Building Agility into Benefit-Risk Assessment

## Weighting Configuration Panel

Weighting Configuration

Method: Swing weighting ▼

- ✓ Swing weighting
- Direct weighting
- Analytical hierarchy

✓ Total: 100%

## Weighting Configuration

Select the appropriate weighting configuration

## Item Weighting Adjustment

● BENEFIT (52%)

HiSCR (Hidradenitis Suppurativa Clinical Response) 13%

Abscess and Inflammatory Nodule (AN) Count 13%

Draining Fistula Count 13%

Flare Rate 13%

## Item Weighting

Decide each benefit & risk item's weight

## Generate BR Analysis

Generate and review the automated results before any human adjustments

## Report Generation Button

RISKS

Save Draft Continue to Display →

# Real-Time Update Management

Our platform can generate up-to-date benefits & risks evaluations to:

- Access all relative evidence
- Evaluate uncertainty in potential benchmarks

Master & Share Real-Time BRA Evolutions

### Benefits

|                                             |                                                 |                 |
|---------------------------------------------|-------------------------------------------------|-----------------|
| Reduction of Stroke<br><span>benefit</span> | Actual benefit<br>Uncertainty: <span>LOW</span> | <span>XX</span> |
| Ischemic Stroke<br><span>benefit</span>     | Uncertain<br>Uncertainty: <span>HIGH</span>     | <span>XX</span> |

### Risks

Collapse all

▼

EYE DISORDERS

0 selected

☐

Uveitis

NEW

Serious

Grade 3-4

Rare

+6 more

>

# Our BRA Platform Manages Safety Team Priorities & Workflow

## BRAT WORKFLOW

- ✓ Step 0: Configuration
- ✓ Step 1: Context
- ✓ Step 2: Outcomes Identification
- ✓ Step 3: Data Analysis
- ✓ **Step 4: Customize**
- ✓ Step 5: Weighting
- ✓ Step 6: Display

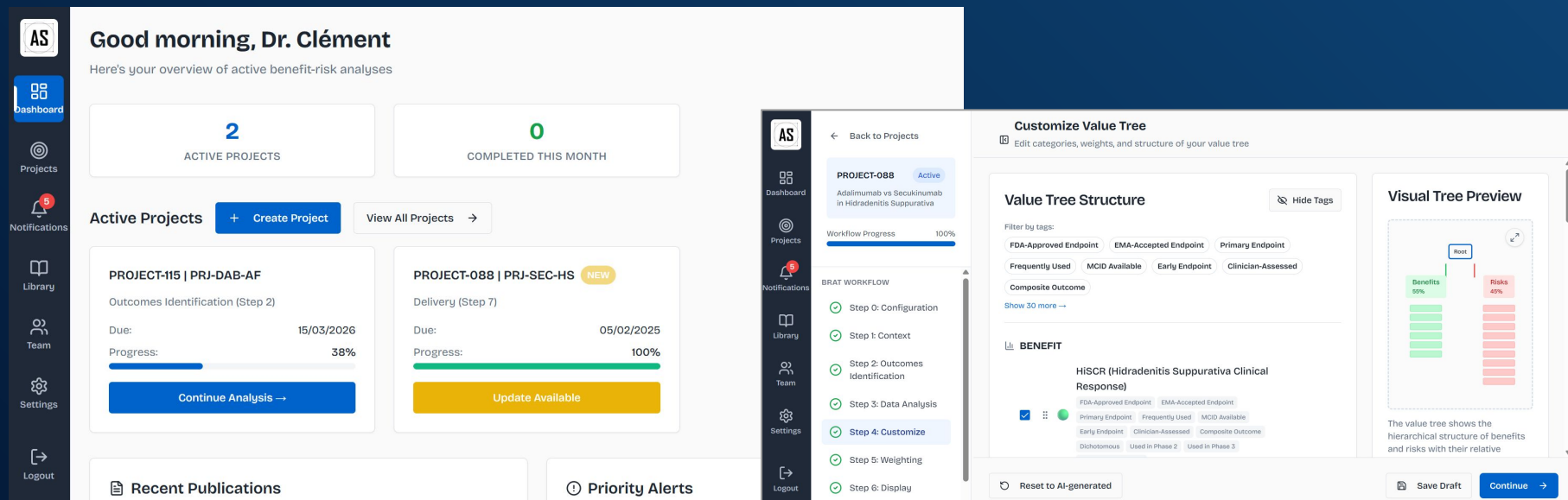
*Structured  
& proven  
BRAT  
methodology*

Customize your BRA by having our engineers :

- Insert additional data sources
- Insert protocol design
- Select the intervention for benchmarking
- Select the appropriate outcomes
- Personalise your BR value tree and related visualisations
- Review AI general synthesis
- Generate intermediary studies

ArcaScience offers Human Oversight for Every Report

# BRA Methodology Implementation by Design



**Good morning, Dr. Clément**  
Here's your overview of active benefit-risk analyses

**2** ACTIVE PROJECTS  
**0** COMPLETED THIS MONTH

**Active Projects** [+ Create Project](#) [View All Projects →](#)

**PROJECT-115 | PRJ-DAB-AF**  
Outcomes Identification (Step 2)  
Due: 15/03/2026  
Progress: 38%  
[Continue Analysis →](#)

**PROJECT-088 | PRJ-SEC-HS** NEW  
Delivery (Step 7)  
Due: 05/02/2025  
Progress: 100%  
[Update Available](#)

**Recent Publications** **Priority Alerts**

**BRAT WORKFLOW**

- Step 0: Configuration
- Step 1: Context
- Step 2: Outcomes Identification
- Step 3: Data Analysis
- Step 4: Customize**
- Step 5: Weighting
- Step 6: Display

**Customize Value Tree**  
Edit categories, weights, and structure of your value tree

**Value Tree Structure** [Hide Tags](#)

Filter by tags:  
☒ FDA-Approved Endpoint  
☐ EMA-Accepted Endpoint  
☐ Primary Endpoint  
☐ Frequently Used  
☐ MCID Available  
☐ Early Endpoint  
☐ Clinician-Assessed  
☐ Composite Outcome  
[Show 30 more →](#)

**BENEFIT**

**HISCR (Hidradenitis Suppurativa Clinical Response)**

☒ FDA-Approved Endpoint  
☐ EMA-Accepted Endpoint  
☐ Primary Endpoint  
☐ Frequently Used  
☐ MCID Available  
☐ Early Endpoint  
☐ Clinician-Assessed  
☐ Composite Outcome  
☐ Dichotomous  
☐ Used in Phase 2  
☐ Used in Phase 3

[Reset to AI-generated](#) [Save Draft](#) [Continue →](#)

**Visual Tree Preview**

The value tree shows the hierarchical structure of benefits and risks with their relative

Reduce BRA Project Time by 80%<sup>1</sup>

<sup>1</sup>: independent BRA experts evaluation, 2025

# With Vs. Without BRA Platform Use

| Activities          | Manual BRA                                     | Assisted BRA                                      |
|---------------------|------------------------------------------------|---------------------------------------------------|
| Literature review   | Months                                         | Instant                                           |
| Benchmarking setup  | Only one (too time-consuming)                  | Unlimited change                                  |
| Workflow management | In-person & emails                             | Team shares access to platform                    |
| BRA adjustment      | Every parameter change generate workload       | Instant impact visualisation                      |
| Outcomes            | High standards require extensive manual effort | Automated excellence:<br>Quality without the wait |

The First End-to-End AI Platform for Benefit-Risk Analysis

# Which collaboration model fits with your needs?

|                                              | Benefits                                                                                                             | Pricing policy                                     |
|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| <b>1 - Autonomous access to the platform</b> | Save time and CRO budget by autonomously generating the Benefit-Risk Assessment (BRA)                                | Price based on the number of BRA project slots     |
| <b>2 - Managing services</b>                 | Work with a single point of contact to access the platform, which includes all integrated data related to your needs | Price based on the number of Benefit-Risk Analysis |
| <b>3 - Substudy automated reports</b>        | Order any report and receive the analysis, along with the related dataset, within one or two days                    | Price based on the number of analysis and topic    |

Contact us to get your tailored proposal





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# Any Questions

Trusted by:



sanofi



Kap•Code



# Unlocking Success through Collaboration



## Goals & means:

- **Reveal the Risk and Benefit** of a monoclonal antibody
- **32 million** open access datasets + **5200 pages** of Sanofi internal data

## Output & take away:

- One predictive **benefit-risk** value tree & cross-comparison between 1st and 2nd gen + A **benefit-risk** competition analysis **27 key inflammatory adverse events** revealed
- **64 key biomarkers** for clinical confirmation

## Impact:

- Literature review time **reduced from 18 months to 2 weeks**

# Unlocking Success through Collaboration



## Goals & means:

- Out of 100 selected drugs available on the market, **find the best candidates for being repurposed to glioblastoma.**
- The first key task is to prove if there is a crossing of the BBB (Blood Brain Barrier), induced by these drugs.

## Output & take away:

- **1 dataset showcasing the safety datapoints** directly connected to the identification of the crossing of the BBB (eye deficiency, headache, earing issue, etc.)

## Impact:

- **3 drugs were found** to be compatible with glioblastoma, **2 are now pushed in phase 2 of clinical trials**